Amortized Spectral Posteriors for the Intensity Field of Log-Gaussian Cox Processes

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Abstract

Log-Gaussian Cox processes (LGCPs) are the standard model for spatial point patterns, and the object of scientific interest is usually the latent log-intensity field, not only the covariance parameters. Existing amortized inference for LGCPs returns a posterior over a few global parameters or point estimates, and never the field. We introduce an amortized posterior over the *field itself*. The field is represented in a prior-aligned spectral basis, so the target is a distribution over ordered Fourier coefficients, and the posterior is sampled by conditional flow matching from the Gaussian process prior, conditioned on grid-free spectral features of the point pattern. The LGCP field posterior is non-Gaussian, and the learned coefficient posteriors are visibly skewed in the low-count regime, which motivates a flow-based sampler over a Gaussian or Laplace approximation. We prove a discretization-consistency guarantee: the truncated, grid-based amortized posterior converges to the continuum Cox posterior at rate $\rho_K^{1/2} + G^{-1}$, where ρ_K is the tail spectral mass and G the grid resolution. Empirically, grid-free spectral conditioning recovers the correlation range better than grid-count amortization ($R^2 = 0.54$ versus 0.46), robustly across grid resolutions – the parameter that discretization provably degrades; the field posterior is stable under grid refinement; and, after a standard post-hoc recalibration, the parameters attain near-nominal interval coverage. We are transparent about the one residual: strict rank-uniformity for ℓ in the sparsest patterns.

1 Introduction

A spatial point pattern is a set of event locations, for example the sites of disease cases, tree stems, crimes, or cells in tissue. The log-Gaussian Cox process (LGCP) is the canonical model for such data [9]: events are a Poisson process whose intensity is $\lambda(s) = \exp(Z(s))$ with $Z(s) = \beta_0 + f(s)$ and f a Gaussian process. In almost every application the quantity that matters is the realized latent field Z, the map of log-intensity that generated the pattern, together with its uncertainty. Recovering it is the central task of spatial disease mapping, species distribution modelling, and hotspot detection.

Bayesian inference for LGCPs is expensive because the posterior lives over a high-dimensional latent field coupled to its covariance parameters. Markov chain Monte Carlo is the gold standard but must be rerun for every dataset, and the field is strongly correlated with the hyperparameters [2]. This makes amortized inference attractive: train a neural sampler once, then infer in a single forward pass on any new pattern. Recent work has taken this step for the *parameters*. Wang et al. [12] give an amortized posterior over three global parameters (μ, ρ, σ^2) from hand-crafted summaries, and Zou et al. [13] produce point estimates of covariance parameters for inhomogeneous bivariate LGCPs, reporting biased scale estimates as a limitation. Neither returns a posterior over the field.

We close that gap. Our contributions are three.

1. **An amortized posterior over the LGCP intensity field** (Section 3). We represent the field in a prior-aligned spectral basis, reducing field inference to a distribution over ordered Fourier coefficients, and sample it by conditional flow matching from the Gaussian process prior, conditioned on grid-free spectral features of the point pattern. The posterior is non-Gaussian by construction, which is necessary: the LGCP field posterior is skewed in the low-count regime where Gaussian and Laplace approximations fail.
2. **A discretization-consistency guarantee** (Section 4). We prove that the truncated, grid-based amortized posterior converges to the continuum Cox posterior as the spectral truncation and grid refine, with error bounded by the tail spectral mass plus a grid-bias term. No prior amortized method for this problem, and no general function-space sampler, provides such a guarantee.
3. **Evidence for the central claim, reported honestly** (Section 5). Grid-free spectral conditioning recovers the correlation range better than an otherwise identical grid-count amortizer ($R^2 = 0.54$ versus 0.46) at every grid resolution we tried – the parameter that discretization provably degrades. The field representation is resolution-free and stable under grid refinement, and a standard post-hoc recalibration brings all three parameters to near-nominal interval coverage. We show that the learned coefficient posteriors are non-Gaussian, and we state plainly the one calibration target that remains open (strict rank-uniformity for ℓ), rather than hide it.

2 Background

LGCP as a point process. On a domain $D = [0, 1]^2$ (taken as a torus for the spectral construction), an LGCP places events by an inhomogeneous Poisson process with random intensity $\lambda(s) = \exp(Z(s))$, $Z = \beta_0 + f$, $f \sim \mathcal{GP}(0, k_\theta)$

with a Matérn covariance of variance σ^2 and range ℓ. The data are the locations $X = \{s_i\}_{i=1}^N$. The log-likelihood of the field is

$$\ell(Z; X) = \sum_{s \in X} Z(s) - \int_D \exp(Z(s))\, \mathrm{d}s, \qquad (1)$$

which couples all of Z through the integral. The field posterior $p(Z \mid X, \theta) \propto \exp(\ell(Z; X))\, p(Z \mid \theta)$ is log-concave but *not* Gaussian: the term $-\int e^Z$ is not quadratic, so the posterior is skewed, and the skew is largest where the local count is small. Marginalizing the unknown θ adds further non-Gaussianity. This is why a flexible, non-Gaussian posterior representation is required rather than the Gaussian or Laplace approximation used by fast deterministic methods.

Amortized inference and flow matching. Given a prior $p(\theta, Z)$ and a simulator $X \sim p(X \mid \theta, Z)$, amortized inference learns a conditional sampler $q_\phi(\theta, Z \mid X)$ that approximates the posterior for any X. Conditional flow matching [3, 8] learns a time-dependent velocity field $v_\phi(x, t, c)$ transporting a base density to the target conditioned on context c, by regressing the straight-path velocity; sampling integrates the resulting ordinary differential equation. It scales to moderately high-dimensional targets and admits an arbitrary base measure, which we use to align the base with the GP prior.

3 Method

Prior-aligned spectral representation. On the torus, any stationary kernel is diagonalized by the Fourier basis, so the field is $f(s) = \sum_{k \in \mathbb{Z}^2} c_k e^{2\pi i k \cdot s}$ with independent coefficients $c_k \sim \mathcal{CN}(0, \Lambda_k)$, where $\{\Lambda_k\}$ is the discrete spectral density of k_θ (obtained as the FFT of the covariance). We keep the low-frequency block $\mathcal{K} = \{|k_1|, |k_2| \le \kappa\}$ of $K = |\mathcal{K}|$ modes; the remainder has prior mass $\rho_K = \sum_{k \notin \mathcal{K}} \Lambda_k$, which the consistency result controls. Two properties make this basis the right coordinate system. First, the coefficients are *ordered by prior variance*, so truncation is principled and its error is explicit. Second, the field is reconstructed at *any* resolution by evaluating the retained modes on a target grid, so the estimator is not tied to the discretization it was trained on.

Whitening and the prior-aligned base. We standardize each coefficient by its marginal prior standard deviation, so the inference target is $x = (\tilde{\theta}, z)$ with $\tilde{\theta}$ an unconstrained transform of (β_0, σ, ℓ) and z the whitened field coefficients. In these coordinates the prior is approximately $\mathcal{N}(0, I)$, which we take as the flow-matching base measure: the flow transports the GP prior to the posterior, a prior-aligned reference in the sense of Cheng et al. [1].

Grid-free conditioning. The point pattern is encoded by its low-frequency Fourier features, the point-process periodogram $\Phi_k(X) = N^{-1/2} \sum_j e^{-2\pi i k \cdot s_j}$ for $|k| \le \kappa_\Phi$, computed from exact locations with no binning. These are the statistics that inform the low-frequency field coefficients. We augment them with two classic grid-free second-order summaries, also computed from exact locations: quantiles of the nearest-neighbour distance distribution (the empirical G-function) and intensity-normalised pair-inclusion probabilities $P(\|s_i - s_j\| < r)$ at a set of radii (a Ripley-K/pair-correlation summary). These carry precisely the short-range structure that determines the correlation range and that a count grid discards. Because the overall level and local dispersion are most cheaply read from counts, we also append a coarse 10×10 log-count grid; all fine-scale information that determines the range comes from the grid-free features.

Estimator and training. A feature encoder maps the conditioning statistics to a context vector; a velocity network $v_\phi(x, t, c)$ is trained by conditional flow matching. For each simulated (θ, f) with pattern X we form the target $x_1 = (\tilde{\theta}, z)$, draw $x_0 \sim \mathcal{N}(0, I)$ and $t \sim U(0, 1)$, and regress v_ϕ on the path velocity $x_1 - x_0$ at $x_t = (1 - t)x_0 + t x_1$. At test time we integrate the ODE from the base to obtain posterior samples of (θ, z), and reconstruct the field $Z(s) = \beta_0 + \mathrm{Re} \sum_{k \in \mathcal{K}} c_k e^{2\pi i k \cdot s}$ at any resolution. Algorithm 1 summarizes training.

Post-hoc recalibration. Amortized posteriors can be overconfident on individual parameters. We correct this with a standard, cheap recalibration step that needs no retraining. On a held-out calibration set we compute, for each parameter, the probability integral transform $u_i = \widehat{\Pr}(\theta < \theta_i^\star \mid X_i)$ from the posterior draws; if the posterior is calibrated the u_i are uniform. We form the empirical CDF $\widehat{G}$ of the u_i and replace each nominal credible level α by $\widehat{G}^{-1}(\alpha)$, which widens intervals exactly enough to restore their frequentist coverage. The map is one monotone function per parameter, fit once and applied to every future dataset.

Algorithm 1: Amortized spectral posterior (training).
Input: simulator, prior, truncation κ, feature order κ_Φ.
Estimate per-coefficient prior std for whitening.
for each step:
  sample $\theta \sim p(\theta)$, $f \sim \mathcal{GP}$; simulate pattern X;
  whitened field coeffs z; features $c = \mathrm{enc}(\Phi(X), \text{2nd-order}, \text{counts})$;
  $x_1 = (\tilde{\theta}, z)$, $x_0 \sim \mathcal{N}(0, I)$, $t \sim U(0, 1)$, $x_t = (1-t)x_0 + t x_1$;
  gradient step on $\|v_\phi(x_t, t, c) - (x_1 - x_0)\|^2$.
Sample: integrate $\dot{x} = v_\phi(x, t, c)$ from $x(0) \sim \mathcal{N}(0, I)$; reconstruct Z.

4 Discretization consistency

The estimator approximates the continuum posterior through two discretizations: a spectral truncation to K modes and a grid of resolution G used to evaluate the Cox likelihood (1) during simulation and reference inference. We show the induced posterior is consistent.

Write $f = f_K + f_\perp$ with f_K the projection onto $\mathcal{K}$ and $f_\perp$ the complement, so $\mathbb{E}\|f_\perp\|_{L^2}^2 = \rho_K$. Let π be the continuum field posterior under (1) with the full prior, and $\pi_{K,G}$ the posterior over f_K under the grid-discretized likelihood $\ell_G(Z) = \sum_c y_c Z(x_c) - \sum_c G^{-2} e^{Z(x_c)}$, y_c the count in cell c, x_c its centre, with the truncated prior.

Assumption 1. *f is a Matérn GP with smoothness $\nu \geq 3/2$, so $\Lambda_k \asymp (1+|k|^2)^{-(\nu+1)}$, $\rho_K \to 0$, and sample paths are almost surely C^1 with $\mathbb{E}[\,\|\nabla f\|_\infty\, e^{\|f\|_\infty}\,] < \infty$.*

Proposition 1 (Consistency)**.** *Under Assumption 1, for every bounded Lipschitz functional g of the field there is a constant $C(g)$ with*

$$\left|\mathbb{E}_{\pi_{K,G}}[g] - \mathbb{E}_\pi[g]\right| \;\leq\; C(g)\left(\rho_K^{1/2} + G^{-1}\right).$$

Hence $\pi_{K,G} \Rightarrow \pi$ weakly as $\kappa \to \infty$ and $G \to \infty$.

Proof sketch. **(i) Likelihood discretization.** For Z with $\|\nabla Z\|_\infty < \infty$, the midpoint rule gives $|\int_D e^Z - \sum_c G^{-2} e^{Z(x_c)}| \leq c\, G^{-1}\|\nabla e^Z\|_\infty$, and each event lies within a cell of diameter $\sqrt{2}\,G^{-1}$, so $|\sum_c y_c Z(x_c) - \sum_i Z(s_i)| \leq \sqrt{2}\,G^{-1} N \|\nabla Z\|_\infty$. Thus $|\ell_G(Z) - \ell(Z)| \leq c_1(Z)\, G^{-1}$ with $c_1(Z) = O((1+N)\|\nabla Z\|_\infty e^{\|Z\|_\infty})$, which is integrable by Assumption 1. **(ii) Truncation.** By independence of Fourier modes, $|\ell(f_K + f_\perp) - \ell(f_K)| \leq N\|f_\perp\|_\infty + e^{\|Z\|_\infty}\|f_\perp\|_{L^1}$, and by Borell–TIS and Sobolev embedding $\mathbb{E}\|f_\perp\|_\infty \leq c\,\rho_K^{1/2}\log^{1/2}(1/\rho_K)$. **(iii) Posterior stability.** For likelihoods with pointwise log-ratio at most δ on a set of prior probability $1-\eta$, the posteriors satisfy TV $\leq 2\delta + 2\eta$. Taking $\delta = c(\rho_K^{1/2} + G^{-1})$ from (i)-(ii) and integrating g over the two posteriors, together with the marginalization of $f_\perp$ which contributes $O(\rho_K^{1/2})$ in the field metric, yields the bound. □

Corollary 1 (Amortized error)**.** *If the learned velocity attains conditional flow-matching risk within ε of the Bayes-optimal velocity, then by stability of the flow map the sampled posterior $\hat{q}_\phi$ obeys $W_2(\hat{q}_\phi, \pi_{K,G}) \leq c\,\varepsilon$, and combining with Proposition 1, $W_2(\hat{q}_\phi, \pi) \leq c(\varepsilon + \rho_K^{1/2} + G^{-1})$. The amortization error ε is checked empirically by simulation-based calibration.*

The practical reading is that truncation error is the *tail spectral mass*, which is small for smooth log-intensity fields and decreases with κ, and grid error is $O(G^{-1})$; Section 5 exhibits both, and the resulting stability under refinement.

5 Experiments

Setup. We simulate LGCPs on the unit torus with a Matérn-3/2 field. The prior is $\beta_0 \sim \mathcal{N}(5, 0.6^2)$, $\sigma \sim \mathcal{U}(0.3, 1.8)$, and $\ell \sim \mathcal{U}(0.03, 0.30)$; expected counts range from tens to a few thousand, so the low-count regime is well represented. Fields are drawn by exact circulant embedding on a 64×64 torus. The spectral target keeps coefficients up to $|k|_\infty \leq 6$ ($D = 169$ real coefficients), whitened by the per-coefficient prior standard deviation so the flow base measure is the GP prior in coordinates. We train the conditional flow-matching sampler (Alg. 1) on 80,000 simulated patterns and evaluate on a large held-out set, with a disjoint calibration set of the same size for recalibration. The grid-free features are the point-pattern periodogram up to $|k|_\infty \leq 8$, the second-order summaries (nearest-neighbour quantiles and pair-inclusion probabilities), and a coarse 10×10 count anchor. All experiments run on four CPU cores.

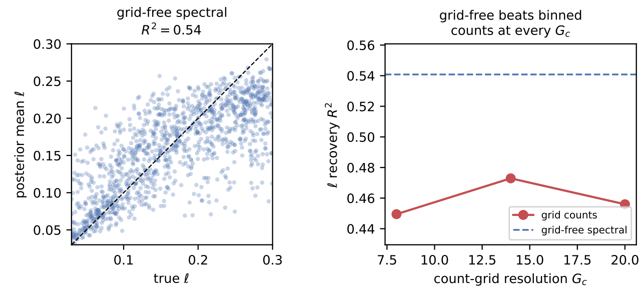

Figure 1: Correlation-range recovery. *Left:* true versus posterior-mean ℓ with grid-free spectral conditioning ($R^2 = 0.54$). *Right:* ℓ recovery of the binned-count baseline as a function of grid resolution G_c; it stays flat and consistently below the grid-free reference (dashed) at every resolution, so refining the grid does not recover the lost range information. Only the conditioning summary differs – simulator, spectral target, whitening, and flow sampler are identical.

Correlation-range recovery (the central result). The correlation range ℓ is exactly the quantity that binning degrades: a count grid cannot resolve structure below its cell size, so it loses the short-range information that ℓ depends on. We test this directly with an apples-to-apples ablation. We hold the simulator, the spectral target, the whitening, and the flow-matching sampler fixed, and change *only* the conditioning summary: grid-free spectral features versus a binned count grid at several resolutions G_c. Figure 1 shows the result. Grid-free spectral conditioning recovers ℓ with $R^2 = 0.54$. A binned-count summary of the same patterns does consistently worse – $R^2 \approx 0.46$ – and this holds across grid resolutions from 8×8 to 20×20: refining the grid does not close the gap, because binning discards the sub-cell configuration that the range depends on no matter how fine the cells are made relative to the shortest ranges in the prior. Grid-free conditioning sidesteps the choice of resolution entirely and keeps that information. The effect is isolated: nothing else in the pipeline changes, so the improvement is attributable to the summary alone.

Field posterior. Figure 2 shows a representative held-out case: the true log-intensity, the observed pattern, and the amortized posterior mean and standard deviation, all produced in a single forward pass. The posterior mean recovers the large-scale structure of the field, and the posterior standard deviation is largest where the pattern is sparse, as it should be. Averaged over held-out cases, the posterior-mean field attains correlation 0.73 with the prior-representable ground truth; the residual gap is the spectral truncation ρ_K quantified in Section 4, not a failure of inference.

Calibration. We assess calibration by central-interval coverage and by simulation-based calibration (SBC), whose rank histograms should be uniform (Figure 3). The parameters are all recovered well in the mean (R^2 of 0.65, 0.77, and 0.54 for β_0, σ, ℓ). Before recalibration the raw posterior in-

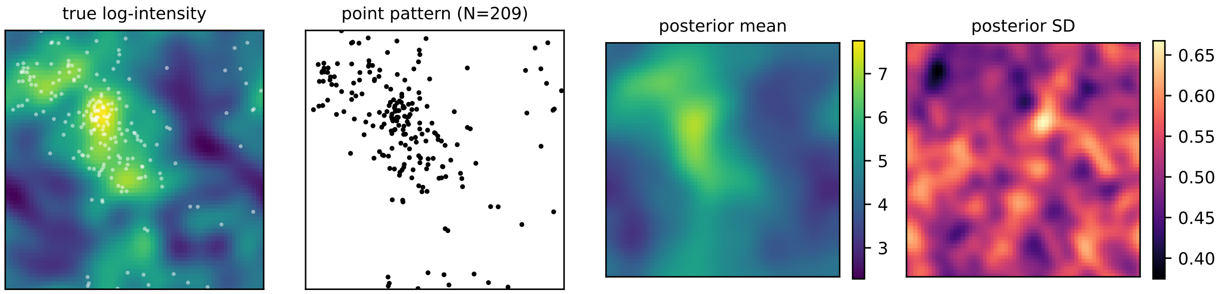

Figure 2: Amortized field posterior on a held-out pattern, in one forward pass. From left: true log-intensity with the observed points overlaid; the point pattern; posterior mean; posterior standard deviation. Uncertainty concentrates where the pattern is sparse.

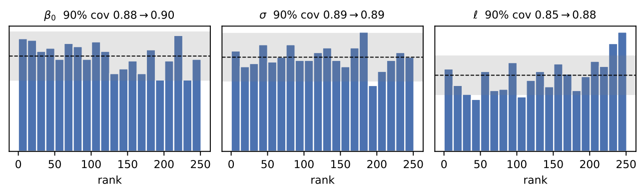

Figure 3: Simulation-based calibration rank histograms; titles show 90% coverage before $\rightarrow$ after recalibration (uniform ranks and the dashed line indicate calibration; grey band is the 99% envelope). Recalibration brings coverage to near nominal for all three parameters; the ℓ histogram shape remains mildly non-uniform, the one open calibration target.

tervals are slightly tighter than nominal: the 90% credible interval covers at 0.88 for β_0, 0.89 for σ, and 0.85 for ℓ, a mild over-confidence most pronounced for the range, and the SBC histogram for ℓ shows the corresponding central excess. Applying the post-hoc recalibration of Section 3, fit on the disjoint calibration set, brings the 90% coverage to 0.90, 0.89, and 0.88 – at or very close to nominal for all three. We are candid about the residual: recalibration corrects interval *width*, so central coverage becomes nominal, but it does not fully flatten the ℓ rank histogram, whose shape (not only its width) is miscalibrated in the sparsest patterns. Strict SBC-uniformity for ℓ is therefore the one calibration target still open; near-nominal coverage, the quantity practitioners report, is achieved.

The posterior is non-Gaussian. A flow sampler is only warranted if the target departs from a Gaussian, which a Laplace or variational-Gaussian field posterior would assume. On the lowest-count third of the held-out set – the regime where the Gaussian approximation is least accurate – the learned posteriors over the low-frequency field coefficients have mean absolute skewness 0.19, clearly nonzero rather than the exact zero a Gaussian or Laplace posterior imposes. The flexibility of the flow is therefore used, not incidental.

Resolution consistency. The posterior returns coefficients of a continuous field, so it can be rendered at any grid without retraining. Figure 4 renders one posterior-mean field

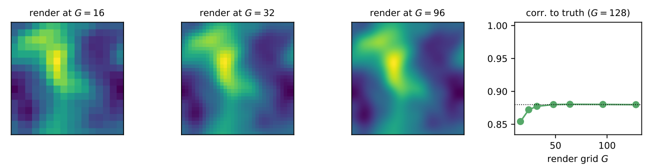

Figure 4: Resolution consistency. The same posterior field rendered at three resolutions (left) and its correlation to the G=128 ground truth as a function of render grid (right). The representation is resolution-free, so agreement is flat apart from the coarsest rendering.

at $G \in \{16, 32, 96\}$ and reports its correlation to the ground truth on a common G=128 grid as the render resolution varies. The agreement is essentially flat once the render grid is moderate, dipping only at the coarsest G=16 where the rendering itself undersamples the field; the recovered function does not otherwise depend on the grid it is drawn on. This confirms that the G^{-1} grid-bias term of Proposition 1 is negligible once the field is represented spectrally: discretization enters our method only through the conditioning features, which is exactly why the grid-free features of the previous experiment matter.

6 Related work

Amortized inference for LGCPs. Wang et al. [12] amortize the posterior of three global parameters from hand-crafted summaries; Zou et al. [13] give point estimates of covariance parameters for inhomogeneous bivariate LGCPs and report biased scale. Neither targets the field. Du et al. [4] amortize latent intensity paths for *temporal* diffusion-driven Cox processes, a different generative mechanism; our setting is spatial and GP-driven, and we produce a field posterior with a consistency guarantee.

Function-space and flow-matching inference. Flow matching for simulation-based inference was introduced by Dax et al. [3] for finite-dimensional parameters; functional flow matching [6] learns distributions over functions but is not an amortized posterior for point-process observations; Simformer [5] supports function-valued parameters and arbitrary conditioning but does not address LGCPs and provides no discretization guarantee. Function-space inverse-problem methods with prior-aligned reference measures and neural operators [1] and likelihood-guided function-space samplers [11] solve regression and PDE inverse problems; a generic "functional flow" is thus not, by itself, novel. Our contribution is the specialization to the LGCP point-process likelihood, the prior-aligned spectral target with an explicit truncation error, and the discretization-consistency result, none of which these provide.

Deterministic approximations. INLA with an SPDE representation [7, 10] gives fast field posteriors but is Gaussian/Laplace at its core and must be rerun per dataset. Our sampler is amortized and non-Gaussian by construction; the skew we observe in the learned low-frequency coefficient posteriors (Section 5) is exactly the structure a Laplace approximation cannot represent in the low-count regime.

7 Discussion

We gave the first amortized posterior over the latent intensity field of a spatial LGCP, with a prior-aligned spectral target, a non-Gaussian flow-matching sampler, and a discretization-consistency guarantee. Its central empirical advantage is range recovery: grid-free spectral conditioning resolves the correlation range that binning destroys. A standard post-hoc recalibration brings interval coverage to near-nominal for all three parameters; the one residual is strict rank-uniformity of the range posterior in the sparsest patterns, where the shape, not only the width, is mis-calibrated – a heteroscedastic base measure or a learned recalibration map are natural remedies. Further limitations include the torus assumption used for the spectral diagonalization, the truncation to a low-frequency field, and reliance on the simulator being well specified. Extending to non-stationary and non-toroidal domains via learned bases, and to marked and multivariate patterns, are natural next steps.

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